

The Heath–Jarrow–Morton Framework

10.1 Model the whole curve, not one point on it

The insight behind Heath, Jarrow and Morton’s 1992 framework is a change of what gets modelled. Instead of specifying dynamics for a single number, $r(t)$, and deriving the rest of the curve from it, HJM directly specifies dynamics for the *entire instantaneous forward curve* $f(t, T)$ (recall from Chapter 2: $f(t, T) = -\partial_T \log P(t, T)$), simultaneously, for every T , as a single infinite-dimensional stochastic object evolving in t .

Key result: the HJM setup

For each fixed maturity T , the instantaneous forward rate evolves as

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T) dW(t)$$

under the real-world measure, where $\alpha(t, T)$ is a drift and $\sigma(t, T)$ a volatility, both permitted to depend on the entire history of the curve up to t . The remarkable result HJM proved is that *once the volatility function $\sigma(t, T)$ is chosen, the drift $\alpha(t, T)$ is no longer free* — no-arbitrage forces it to be a specific, determined function of σ alone.

10.2 The HJM drift condition

This is the single most important result in the framework, and it is worth stating precisely because its structure — a volatility choice determines a mandatory drift — reappears, essentially unchanged in spirit, as the central mechanic of the BGM model in Chapter 13.

Key result: the HJM no-arbitrage drift condition

Under the risk-neutral measure, no-arbitrage requires

$$\alpha(t, T) = \sigma(t, T) \int_t^T \sigma(t, u) du$$

That is, the drift of the instantaneous forward rate at maturity T is fixed, exactly, by its own volatility integrated against the volatility of every forward rate between t and T . Choose *any* volatility structure $\sigma(t, T)$, plug it into this formula, and the resulting model is automatically arbitrage-free and automatically consistent with today’s observed curve (since the curve enters only as the model’s initial condition $f(0, T)$, not through any fitted parameter).

This is a genuinely elegant unification: every specific short-rate model from Chapter 9 can be recovered as a special case of HJM for a particular choice of $\sigma(t, T)$ (Hull–White, for instance, corresponds to an exponentially decaying, deterministic σ). HJM is not a competitor to short-rate models; it is the general theory that contains them, and that fact alone explains why it was received, on publication, as a genuine breakthrough in term-structure modelling.

10.3 Why HJM, despite this, is hard to use directly

Having unified the field theoretically, HJM created a new practical problem, and it is worth being honest about exactly what that problem is, because it is precisely what BGM was invented to fix.

PITFALL

HJM is generically non-Markovian. Because the drift at maturity T depends on an integral of volatility across the whole curve up to T , and because $\sigma(t, T)$ is allowed to depend on the curve's own history, the resulting short-rate process $r(t) = f(t, t)$ is, for a generic choice of σ , path-dependent in a way that cannot be reduced to a small number of state variables. Concretely: pricing a derivative under a generic HJM model usually requires simulating the *entire* forward curve, at every time step, which is computationally expensive, and building an efficient tree or lattice — straightforward for a one-dimensional Markovian short-rate model — is generally not possible at all.

PITFALL

$f(t, T)$ is not something anyone trades. An instantaneous forward rate is a mathematical idealisation — the limit of a forward rate over an infinitesimally short period. No market participant quotes or trades $f(t, T)$ directly; what trades is the LIBOR- or SOFR-style *simple* forward rate over a finite period (Chapter 2). This means calibrating an HJM volatility function $\sigma(t, T)$ to market cap and swaption volatilities requires yet another non-trivial translation step, similar in spirit to the short-rate translation problem of Chapter 9, just dressed in more elegant mathematics.

ON THE DESK

HJM's continuing legacy on a trading desk is less as a model that gets implemented directly, and more as the theoretical language in which no-arbitrage arguments for *any* term-structure model get made and checked — including BGM's own drift condition in Chapter 13, which is most cleanly derived by recognising it as HJM's drift condition, applied to simple rather than instantaneous forward rates. When a quant says a proposed volatility structure “isn't HJM-consistent,” they mean it fails to satisfy some version of the drift condition of this chapter, and would therefore permit arbitrage if used naively.

10.4 The gap BGM closes

The unresolved tension at the end of this chapter is exactly the motivation for the next one: HJM gives a completely general, provably arbitrage-free recipe for building a term-structure model, but its natural state variable ($f(t, T)$) is unobservable and its generic implementation is computationally difficult. What if the same no-arbitrage logic could be applied not to the instantaneous forward rate, but directly to the *simple, market-quoted* forward rate $L(t; T_i, T_{i+1})$ — the exact object caps and swaptions are actually written on? That question, asked independently by Brace, Gatarek and Musiela and by Miltersen, Sandmann and Sondermann in 1997, is where the next chapter begins.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. State the HJM no-arbitrage drift condition and explain, in words, what it says: volatility choice determines drift.
2. Explain why HJM is described as a unifying framework that contains every short-rate model as a special case.
3. Name the two practical obstacles — non-Markovianity and the unobservability of $f(t, T)$ — that keep HJM from being used directly for calibration to the cap/swaption market.

